# ISI
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(i) Determine the profit maximizing price and quantity of the monopolist.

(ii) Suppose that the patent expires at a certain point of time after which any new drug company can enter the market and produce Drug $X$, facing the same cost function $C = 25 + X^2$. What will be the competitive equilibrium industry output and price? How many firms will there be in the market?

8. Consider an economy where there are two commodities 1 and 2 and two consumers A and B. Assume that both consumers have identical preferences described by the utility function $U = x_1 x_2$ where $x_1$ and $x_2$ denote the quantities of commodities 1 and 2 respectively that are consumed. Consumer A has an endowment of 1 unit of commodity 1 and none of commodity 2 while consumer B has 1 unit of commodity 2 and none of commodity 1.

   (i) Describe the set of Pareto-efficient allocations in the economy.

   (ii) What is the competitive equilibrium allocation?

   (iii) Consider now a four-person economy which is the economy described above but where consumers A and B now have a "clone" each. In other words, there are now two consumers who have the same preferences and endowments as Consumer A and similarly for Consumer B. What is the competitive allocation for this economy?

AB



2001

**Test Code : MEI/MEII 2001**

**Syllabus for MEI**

*Matrix Algebra :* Matrices and Vectors, Matrix Operations, Determinants, Non-singularity, Inversion, Cramer's rule.

*Calculus :* Limits, Continuity, Differentiation of functions of one or more variables, Product rule, Partial and total derivatives, Derivatives of implicit functions, unconstrained optimisation (first and second order conditions for extrema of functions of single variable, and several variables), Taylor series, Definite and indefinite integrals standard formulae, integration by parts and integration by substitution, Differential equations.

Constrained optimisation of functions of a single variable.

Theory of sequences and series.

*Linear programming :* Formulation, Statements of primal and dual problems, Graphical solutions.

Theory of polynomial equations (up to third degree).

*Sample Questions MEI (Mathematics) 2001*

1. The sequence $\frac{5n}{n+2}$ is

   (A) unbounded and monotone increasing

   (B) bounded and monotone decreasing

   (C) bounded and monotone increasing

   (D) unbounded and monotone decreasing

2. Let $f : \Re \to \Re$ be defined by

$$f(x) = \begin{cases} x & \text{if } x \geq 1 \\ x^2 & \text{if } x < 1 \end{cases}$$

   Then

   (A) $f$ is continuous but not differentiable at $x = 1$

   (B) $f$ is not continuous but is differentiable at $x = 1$

   (C) $f$ is continuous and differentiable at $x = 1$

   (D) $f$ is neither continuous nor differentiable at $x = 1$

3. The infinite series $\frac{x}{1^2}+\frac{x^2}{2^2}+\frac{x^3}{3^2}+\ldots$ is

(A) convergent if $|x|>1$

(B) convergent if $|x|<1$

(C) divergent if $|x|<1$

(D) none of the above

4. The range of the function $f(x)=\frac{2x^2}{1+3x^2}$ defined over the set of real numbers is

(A) $[0,\infty)$

(B) $[0,\frac{2}{3})$

(C) $[0,\frac{2}{3}]$

(D) $(-\infty,\infty)$

5. The determinant of the matrix $\begin{bmatrix} 2 & 4 & 3 \\ 5 & 10 & 11 \\ 2 & 4 & 10 \end{bmatrix}$ is

(A) 0

(B) 4

(C) -4

(D) 3

6. The inverse of the function $f(x)=\frac{1}{x}, x>0$, is the function

(A) $g(x)=x$

(B) $g(x)=-x$

(C) $g(x)=-\frac{1}{x}$

(D) $g(x)=\frac{1}{x}$

7. $Limit_{n\to\infty}\frac{2n^3-3n}{5n^3+4n^2-2}$ is

(A) -1/2

(B) -3/4

(C) 2/5

(D) 1



8. If $x=\log_{12}18$ and $y=\log_{24}54$, then $xy+5(x-y)$ is

(A) negative

(B) positive but less than 1

(C) 1

(D) 2

9. The linear programming problem

Maximize $x_1+x_2$

Subject to $x_1\geq x_2-1,\ x_2\leq 2+\frac{x_1}{2}$

$x_1, x_2\geq 0$

(A) has no feasible solution

(B) has a unique optimal solution

(C) has more than one optimal solution

(D) has no optimal solution

10. Given $f(16)=16$ and $f'(16)=4$

$Limit_{x\to 16}\frac{\sqrt{f(x)}-4}{\sqrt{x}-4}$ equals

(A) 3

(B) 4

(C) 5

(D) 2

11. The integral $\int\frac{4x^3+2}{x^4+2x}dx$ can be evaluated as

(A) $\frac{x^4+2x}{4x^3+2}$ + constant

(B) $\ln x^4+\ln 2x$ + constant

(C) $\frac{1}{2}\ \ln\ |x^4+2x|$ + constant

(D) $\frac{|x^4+2x|}{|4x^3+2|}$ + constant

12. Let $(a_n)$ be a sequence of real numbers not converging to zero. Then

(A) $(a_n)$ converges to a non-zero number

(B) for any $\epsilon > 0$, there exists N such that $| a_n | > \epsilon$

(C) there exists $\epsilon > 0$ such that for any N, $| a_n | > \epsilon$ for all $n > N$

(D) there exists N such that $| a_n | > \frac{1}{n}$ for all $n > N$

13. $Limit_{x \to \infty} \left(1 + \frac{3}{x}\right)^x$ equals

(A) e

(B) $e^3$

(C) 0

(D) 1

14. The solution of the differential equation $\frac{dx}{dt} + 2x = 6$ with $x(0) = 10$ is

(A) $x = 10e^{-2t} + 3$

(B) $x = 7e^{-2t} + 3$

(C) $x = 8e^{-2t} + 3$

(D) $x = \frac{10}{7}e^{-2t} + 3$

15. Let X, Y, Z be $n \times n$ matrices such that $XT = YZYX = I$ where I is the $n \times n$ identity matrix. Then

(A) $X = Y$

(B) $Y = Z$

(C) $Z = X$

(D) $XZY = YZX$



## *Syllabus and Sample Questions for MEII (Economics) 2001*

### *Syllabus : MEII*

*Microeconomics* : Consumer theory, Producer theory, Market forms, Welfare economics; *Macroeconomics* : National income accounting, simple model of income determination and multiplier; Banking, money supply and inflation. Applicants are expected to have familarity of these topics at the level of Samuelson and Nordhous (Economics), and Lipsey (Positive Economics).

1. Instruction for question numbers 1(a) - (d). For each of the following questions four alternative answers are provided. Choose the answer that you consider to be the most appropriate for a question and write it down in your answer book.

(i) Suppose that a consumer consumes only two goods, $x_1$ and $x_2$. If the income elasticity of $x_1$ is 1.2, then the income elasticity of $x_2$ is

(A) greater than 1.2

(B) less than 1.2 but greater than 1

(C) less than 1

(D) exactly 1.2

(ii) In a standard Keynesian economy, the consumption function is given by $C = 15+Y^{0.5}$ and the sum of autonomous investment and government spending is 585. The equilibrium $Y$ in the economy equals

(A) 600

(B) 570

(C) 625

(D) 900

(iii) For a monopolist, when the marginal cost of production increases for every level of output :

(A) its profit maximizing price falls

(B) its profit maximzing price increases

(C) its profit maximizing price remains unchanged

(D) how its profit maximizing price changes depends on the demand curve

(iv) If the government imposes a 10% tax on a monopolist's profit, then the monopolist will

(A) decrease his output by greater than 10%

(B) decrease his output by 10%

(C) not change his output

(D) decrease his output but by less than 10%

2. Petrol is sold in a competitive market. The market demand for petrol is $Q^D = 100 - 2P_c$ where $P_c$ is the price paid by consumers. The supply curve is given by $Q^s = 3P_s$ where $P_s$ is the price received by suppliers. Assume that the quantities are measure in litres and price is in terms of Rs per litre.

(i) If there are no taxes on petrol, what will be the equilibrium price and quantity?

(ii) Suppose the government imposes a unit tax of Re 1 per litre of petrol. In the new post-tax equilibrium, what price do the consumers pay and what price do the suppliers receive? How much petrol is sold? What is the excess burden of the tax?

3. A consumer consumes only two commodities $x_1$ and $x_2$. Suppose that her utility function is given by $U(x_1, x_2) = \min(2x_1, x_2)$.

(i) Draw a representative indifference curve of the consumer.

(ii) Suppose the prices of the commodities are Rs 5 and Rs 10 respectively while the consumer's income is Rs 100. What commodity bundle will the consumer purchase?

(iii) Suppose that the price of commodity 1 now increases to Rs 8. Decompose the change in the amount of commodity 1 purchased into income and substitution effects.

4. Consider a simple Keynesian model for an open economy where investment and export demand are autonomous. What happens to GDP and aggregate imports if the import propensity increases in this model?



5. Consider the IS-LM model. In the commodity market, let the consumption function be given by $C = a + bY$, $a > 0$, $0 < b < 1$. Investment and government spending are exogenous and given by $I_0$ and $G_0$ respectively. In the money market, the real demand for money is given by $L = kY - gr$, $k > o$, $g > 0$. The nominal money supply is exogenous and equal to $M_0$; thus the real money supply is $M_0/P$. In this question $C, Y, r$ and $P$denote consumption, real GDP, interest rate and the price level respectively.

(i) Set up the IS-LM equations.

(ii) Determine how an increase in the price level would affect real GDP and the interest rate.

6. A price taking firm makes machine tools $Y$ using labour and capital according to the production function $Y = K^{0.25} L^{0.25}$. Labour can be hired at the beginning of every week while capital can be hired only at the beginning of every month. The wage rate and the rental rate of capital are both 10.

Given the above information, the short run (week) total cost of producing $Y$ machine tools is $10Y^4/K^*$ when the amount of capital is fixed at $K^*$. In the long run (month) when both inputs can be varied, the cost of producing $Y$ is $20Y^2$.

(i) At the beginning of the month of January, the firm is making long-run decisions given that the price of machine tools is 400. What is the long run profit maximizing number of machine tools? How many units of labour and capital should the firm hire at the beginning of January?

(ii) Suppose that at the beginning of the third week of January, the price of machine tools increases to 691.20. What is the new short run profit maximizing level of output? How many new labourers will the firm hire for the rest of January?

7. A drug company is a monopoly supplier of Drug X which is protected by a patent. The demand for the drug is $P = 100 - X$ and the monopolist's cost function is $C = 25 + X^2$.